

1

DATA SOURCES

26,000+ candidate images · 11 folders

Kaggle

SCIN

facial_skin_concerns

5 concerns, weak

existing_conditions

3 concerns, weak

skin_issues_v2

review / external eval

ACNE04

blemish ±

FFHQ-Wrinkle

1,000 masks · fine lines

MEMI-DS

+120 dark-spot positives (rejected)

PASCAL VOC 2007

300 non-face OOD

SCIN

review pool · 0 labels created

4 folders

types, clean, rajesh

Apache-2.0 claimed

licence unknown

pending licence

academic

CC BY-NC-SA 4.0

CC BY 4.0

evaluation only

quarantined

rejected

2

AUDIT & CURATION

scripts/audit_vision_dataset.py

Python

OpenCV

SHA-256 EXACT

7,200+

images in duplicate groups

DHASH · HAMMING ≤ 3

14,900+

near-duplicate group members

PRE-AUGMENTED

2,600+

flagged copies

YUNET FACE GATE ≥ 0.9

3,900+

pass strict full-face check

SOURCE REGISTRY DECISION

accepted_academic

partial_review

pending_license

evaluation_only

rejected

4

SPLITS & MANIFESTS

finalize + validate_vision_manifests.py

pandas

Group-aware split ≈ 55 / 20 / 25 — duplicate groups never cross splits. Gold: 600+ / 200+ / 250+.

+ FFHQ-Wrinkle (fine lines only): wrinkle-pixel share in thirds →

absent

unknown

present

MANIFESTS_HUMAN_REVIEWED

validation_report.json

· 2,000+ IMAGES

train 1,200+

val 400+

test 470+

Val & test files kept **byte-identical** across every retrain, so candidate models are compared fairly.

3

HUMAN ANNOTATION

app/annotate_vision_dataset.py

Streamlit

BALANCED REVIEW QUEUE

1,400+ → 1,100+

images accepted as gold

ROUND 2 · 200 TRAIN IMAGES

868

unknowns filled · 5 rejected

PER-IMAGE LABEL SCHEMA

blemishes

dark_spots

redness

visible_pores

fine_lines

each: present 1 · absent 0 · unknown -1

Source label **hidden** during review to avoid anchoring

Outputs gold_labels.csv + provenance.json (SHA-256 of every input file)

5

PREPROCESS & TRAIN

src/model/train_concern.py · batch 32 · seed 42

TensorFlow

Keras

RGB load PIL

→

Resize

224 × 224

→

preprocess_input

EfficientNet

→

Augment

flip · rot 0.03 · contrast 0.1

⇒

EfficientNetB0

ImageNet · no top

→

GAP pool

→

Dropout

0.3

→

Dense

128 ReLU

→

Dense

5 sigmoid

PHASE A

Backbone frozen

Adam 1e-3 · ≤ 12 epochs

PHASE B

Top 30 unfrozen

Adam 1e-5 · ≤ 7 epochs

LOSS

Masked BCE

unknown -1 → weight 0

CHECKPOINT

Val macro ROC-AUC

early stop · LR on plateau

LEARNING CURVE

25 → 100 % gold

150 → 600+ imgs · 2 seeds

6

EVALUATE

src/model/evaluate_concern.py

scikit-learn

Per-concern threshold chosen on **validation** (Youden J); test set used **only for reporting**.

TEST ROC-AUC PER CONCERN

chance = 0.5

Blemishes

0.86

Dark spots

0.58

Redness

0.71

Visible pores

0.60

Fine lines

0.92

TEST MACRO · 582 LABEL DECISIONS

0.79

0.73

0.90

76.3%

F1

ROC-AUC

PR-AUC

pooled acc.

7

DEPLOY

models/concern_human_reviewed

FastAPI

ONNX

SHIPPED ARTEFACTS

skin_concern_pilot.weights.h5

skin_concern_pilot.json labels · 224 px

evaluation.json per-concern thresholds

face_detection_yunet_2023mar.onnx

SKINANALYZER AT RUNTIME

face gate → shared preprocessing → 5 scores vs threshold

present

uncertain ±0.05

absent

Response schema_version 2 + cosmetic-only disclaimer → recommendation engine